NumPy Matrix Operations

Task 14 Report | Level 1, Day 14 | AI & ML Track

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1. Objective

To create two matrices using NumPy and perform addition, subtraction, matrix multiplication and transpose operations, in order to understand the basic matrix operations used in machine learning.

2. Tools Used

Python, NumPy, Jupyter Notebook.

3. Input Matrices

Matrix	Values
A	[[1, 2], [3, 4]]
B	[[5, 6], [7, 8]]

Both matrices were created manually using np.array(), as suggested in the task.

4. Code

```
import numpy as np

A = np.array([[1, 2], [3, 4]])
B = np.array([[5, 6], [7, 8]])

print("Addition:\n", A + B)
print("Subtraction:\n", A - B)
print("Matrix Multiplication:\n", A @ B)
print("Transpose of A:\n", A.T)
print("Transpose of B:\n", B.T)
```

5. Results

Operation	Result
Addition (A + B)	[[6, 8], [10, 12]]
Subtraction (A - B)	[[-4, -4], [-4, -4]]
Matrix Multiplication (A @ B)	[[19, 22], [43, 50]]
Element-wise Multiplication (A * B)	[[5, 12], [21, 32]]
Transpose of A (A.T)	[[1, 3], [2, 4]]
Transpose of B (B.T)	[[5, 7], [6, 8]]

6. Explanation of Matrix Multiplication

In matrix multiplication, each row of the first matrix is multiplied element by element with each column of the second matrix, and the products are summed to give one entry of the result. For example, result[0][0] = (1 x 5) + (2 x 7) = 19, and result[0][1] = (1 x 6) + (2 x 8) = 22. The multiplication is valid only when the number of columns in the first matrix equals the number of rows in the second matrix. Unlike element-wise multiplication (A * B), which multiplies matching positions, matrix multiplication (A @ B) uses dot products of rows and columns, so A @ B is generally not equal to B @ A.

7. Importance in Machine Learning

Datasets are stored as matrices (rows = samples, columns = features). Linear regression and neural network layers are computed as X @ W + b, so matrix operations let models process many samples at once and run efficiently on CPUs and GPUs. Transpose is used widely, for example in the normal equation and in computing gradients.

8. Conclusion

All required operations were performed successfully with NumPy and the outputs were verified by hand for matrix multiplication. This task built a clear understanding of the matrix fundamentals that underpin machine learning algorithms.